

CORE PURE / COMPLEX NUMBERS

De Moivre's Theorem

Questions by topic

18 questions · 180 marks

Questions

Question 1	7 marks	2
Question 2	13 marks	3
Question 3	4 marks	4
Question 4	19 marks	5
Question 5	12 marks	6
Question 6	7 marks	7
Question 7	8 marks	8
Question 8	9 marks	9
Question 9	8 marks	10
Question 10	8 marks	11
Question 11	9 marks	12
Question 12	10 marks	13
Question 13	10 marks	14
Question 14	14 marks	15
Question 15	12 marks	16
Question 16	9 marks	17
Question 17	11 marks	18
Question 18	10 marks	19

1. A complex number z has modulus 1 and argument θ .

(a) Show that

$$z^n + \frac{1}{z^n} = 2 \cos n\theta \quad \text{and} \quad z^n - \frac{1}{z^n} = 2i \sin n\theta, \quad n \in \mathbb{Z}^+ \quad [3]$$

(b) Hence, by expanding $(z + \frac{1}{z})^3$ and $(z - \frac{1}{z})^3$, show that

$$\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \quad [4]$$

2. (a) Using de Moivre's theorem, show that

$$\sin^4 \theta \cos^2 \theta \equiv \frac{1}{32}(\cos 6\theta - 2 \cos 4\theta - \cos 2\theta + 2) \quad [5]$$

(b) By using the substitution $\alpha = (\frac{\pi}{2} - \theta)$, or otherwise, find a similar identity for $\sin^2 \theta \cos^4 \theta$. [3]

(c) Given that $0 < a < \frac{\pi}{2}$ and

$$\int_0^a (\sin^4 \theta \cos^2 \theta + 2 \sin^2 \theta \cos^4 \theta) \, d\theta = \frac{\pi}{32} - \frac{\sqrt{3}}{64}$$

find the exact value of a . [5]

- 3.** Use de Moivre's theorem to find the constants A , B and C in the identity

$$\sin \theta \cos^4 \theta = A \sin \theta + B \sin 3\theta + C \sin 5\theta \quad [4]$$

4. (i) Starting with the result

$$e^{i\theta} = \cos \theta + i \sin \theta$$

show that

(A)

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta \quad [2]$$

(B)

$$\sin \theta = \frac{1}{2i} (e^{i\theta} - e^{-i\theta}) \quad [2]$$

(ii) Using the result in part (i)(A), obtain the values of the constants a , b , c , d and e in the identity

$$\cos 8\theta \equiv a \sin^8 \theta + b \sin^6 \theta + c \sin^4 \theta + d \sin^2 \theta + e \quad [6]$$

(iii) Using the result in part (i)(B), obtain the values of the constants P , Q , R , S and T in the identity

$$\sin^8 \theta \equiv P \cos 8\theta + Q \cos 6\theta + R \cos 4\theta + S \cos 2\theta + T \quad [6]$$

(iv) Hence show that

$$\sin \frac{\pi}{8} = \left(\frac{17 - 12\sqrt{2}}{64} \right)^{\frac{1}{8}} \quad [3]$$

5. (a) Use de Moivre's theorem to show that $\cos 6\theta \equiv (1 - 2 \sin^2 \theta)(16 \sin^4 \theta - 16 \sin^2 \theta + 1)$ [5]

(b) By solving the equation $\cos 6\theta = 0$, deduce that $\sin^2 \left(\frac{\pi}{12} \right) = \frac{2-\sqrt{3}}{4}$. [4]

(c) Hence, or otherwise, write down the exact values of $\sin^2 \left(\frac{5\pi}{12} \right)$, $\sin^2 \left(\frac{7\pi}{12} \right)$ and $\sin^2 \left(\frac{11\pi}{12} \right)$. [3]

6. Let $t = \tan \theta$. By writing

$$\cos \theta + i \sin \theta = \cos \theta (1 + it)$$

and expanding $(1 + it)^6$, use de Moivre's theorem to show that

$$\tan 6\theta = \frac{6 \tan \theta - 20 \tan^3 \theta + 6 \tan^5 \theta}{1 - 15 \tan^2 \theta + 15 \tan^4 \theta - \tan^6 \theta} \quad [7]$$

7. (a) Use de Moivre's theorem to show that

$$\sin 6x = \sin 2x (a \cos^4 x + b \cos^2 x + c)$$

where a , b and c are integers to be determined.

[4]

- (b) Hence solve, for $0 < \theta < \frac{\pi}{2}$,

$$\sin 6\theta = \sin 2\theta \cos 2\theta$$

giving your answers to 3 decimal places.

[4]

8. (a) By considering the binomial expansion of $(z - z^{-1})^6$, where $z = \cos \theta + i \sin \theta$, use de Moivre's theorem to show that

$$\sin^6 \theta = \frac{1}{32} (10 - 15 \cos 2\theta + 6 \cos 4\theta - \cos 6\theta) \quad [5]$$

- (b) Hence, using the substitution $x = \tan \theta$, find the exact value of

$$\int_0^1 \frac{x^6}{(1+x^2)^4} dx \quad [4]$$

9. (a) Use de Moivre's theorem to show that

$$2 \cos 5\theta = 32 \cos^5 \theta - 40 \cos^3 \theta + 10 \cos \theta \quad [4]$$

(b) Hence solve the equation

$$x^5 - 5x^3 + 5x + 1 = 0$$

giving all roots in the form $2 \cos q\pi$, where q is a rational number. [4]

10. (a) If n is a positive integer and $z = \cos \theta + i \sin \theta$, show that

$$z^n + \frac{1}{z^n} = 2 \cos n\theta, \quad z^n - \frac{1}{z^n} = 2i \sin n\theta \quad [2]$$

(b) By considering $\left(z + \frac{1}{z}\right)^3 \left(z - \frac{1}{z}\right)^2$, find constants A , B and C such that

$$\sin^2 \theta \cos^3 \theta = A \cos 5\theta + B \cos 3\theta + C \cos \theta \quad [6]$$

11. (a) By writing $1 + i \tan \theta = \sec \theta (\cos \theta + i \sin \theta)$ and using de Moivre's theorem, show that

$$(1 + i \tan \theta)^4 = \frac{\cos 4\theta + i \sin 4\theta}{\cos^4 \theta} \quad [3]$$

(b) Expand $(1 + i \tan \theta)^4$ and hence show that

$$\cos 4\theta = \frac{1 - 6 \tan^2 \theta + \tan^4 \theta}{(1 + \tan^2 \theta)^2}$$

and

$$\sin 4\theta = \frac{4 \tan \theta (1 - \tan^2 \theta)}{(1 + \tan^2 \theta)^2} \quad [4]$$

(c) Deduce that

$$\tan 4\theta = \frac{4 \tan \theta (1 - \tan^2 \theta)}{1 - 6 \tan^2 \theta + \tan^4 \theta} \quad [2]$$

12. (a) Use de Moivre's theorem to show that

$$\sec 6\theta = \frac{\sec^6 \theta}{32 - 48 \sec^2 \theta + 18 \sec^4 \theta - \sec^6 \theta} \quad [6]$$

(b) Hence obtain the roots of the equation

$$x^6 - 18x^4 + 48x^2 - 32 = 0$$

in the form $\sec(q\pi)$, where q is rational. [4]

13. (a) Use de Moivre's theorem to show that

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta \quad [5]$$

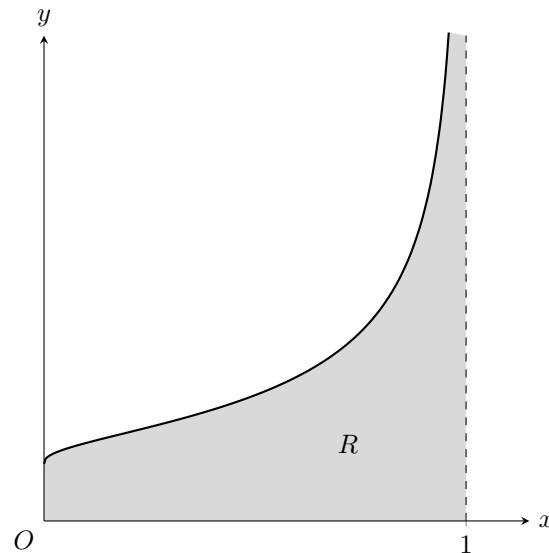
(b) Hence solve for $0 \leq \theta \leq \pi$

$$16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta = \cos 2\theta$$

giving your answers as exact multiples of π . [5]

14. (a) Use de Moivre's theorem to determine constants A , B and C such that

$$\cos^5 \theta \equiv A \cos 5\theta + B \cos 3\theta + C \cos \theta \quad [5]$$



The function f is defined by

$$f(x) = -3 \sin \left(5 \cos^{-1} \left(x^{\frac{1}{6}} \right) \right) - 25 \sin \left(3 \cos^{-1} \left(x^{\frac{1}{6}} \right) \right) - 150 \sin \left(\cos^{-1} \left(x^{\frac{1}{6}} \right) \right), \quad 0 \leq x < 1$$

- (b) Show that

$$f'(x) = \frac{40}{\sqrt{1 - x^{\frac{1}{3}}}} \quad [6]$$

The diagram shows the curve with equation

$$y = \frac{1}{\sqrt{1 - x^{\frac{1}{3}}}}$$

for $0 \leq x < 1$ and the asymptote $x = 1$. The region R is the unbounded region between the curve, the x -axis, the line $x = 0$ and the line $x = 1$.

You are given that the area of R is finite.

- (c) Determine the exact area of R . [3]

15. (i) Use de Moivre's theorem to show that

$$\cos 5\theta \equiv \cos \theta (1 - 12 \sin^2 \theta + 16 \sin^4 \theta) \quad [5]$$

- (ii) Hence determine the range of values of

$$\frac{\cos 5\theta}{\cos \theta}$$

given that $\cos \theta \neq 0$.

[7]

- 16. (a)** Use de Moivre's theorem to show that

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta \quad [5]$$

- (b)** Use the identity given in part (a) to find the 2 positive roots of

$$64x^5 - 80x^3 + 20x + 1 = 0$$

giving your answers to 3 significant figures. [4]

17. (a) Use de Moivre's theorem to show that

$$\sin^2 \theta \cos^2 \theta = \frac{1}{8}(1 - \cos 4\theta) \quad [5]$$

(b) Hence, or otherwise, find the solution of the differential equation

$$\frac{dy}{d\theta} - 2y \tan \theta = \sin^2 \theta$$

for which $y = 0$ when $\theta = \frac{1}{4}\pi$. [6]

18. (a) Use de Moivre's theorem to show that

$$\cos^6 \theta = a \cos 6\theta + b \cos 4\theta + c \cos 2\theta + d$$

where a , b , c and d are constants to be found.

[5]

(b) Hence show that

$$\int_0^{\pi/4} \cos^6 \theta \, d\theta = \frac{11}{48} + \frac{5\pi}{64}$$

[5]